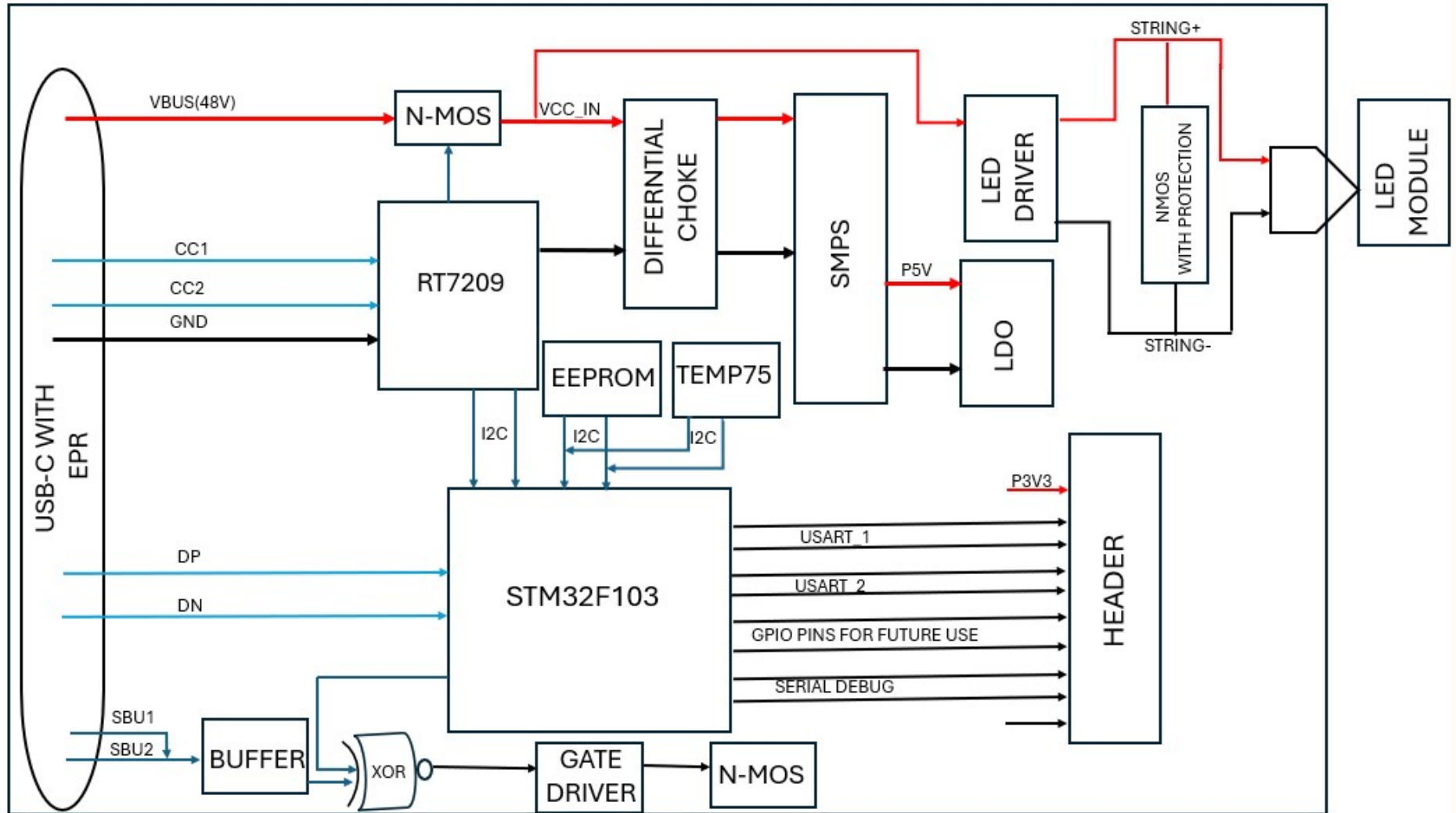


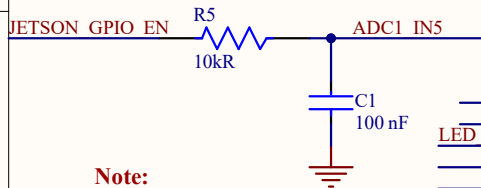
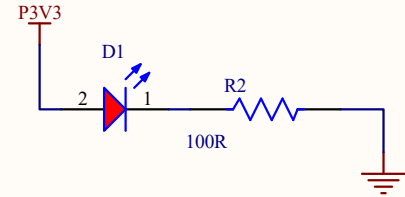
LED DRIVING MODULE ARCHITECTURE



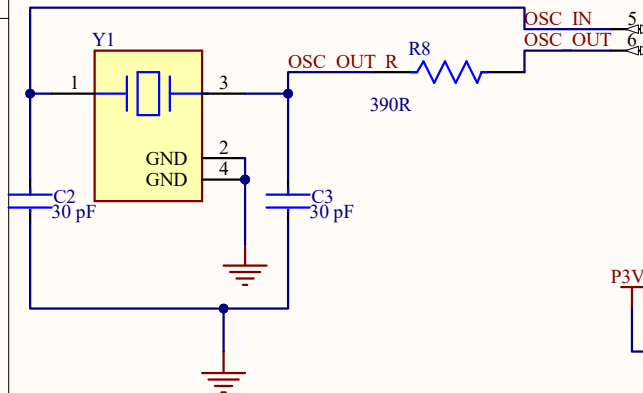
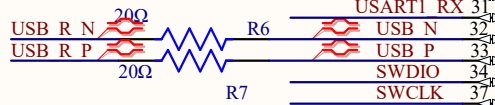
Title		
LED DRIVING MODULE		
Size	Number	Revision
A		V1.0
Date:	6/14/2026	Sheet of
File:	C:\Users\...\BLOCK DIAGRAM.SchDoc	Drawn By: GADILINGAREDDY G M

POWER LED

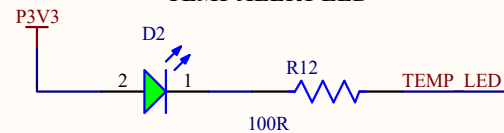
ON BOARD CONTROLLER



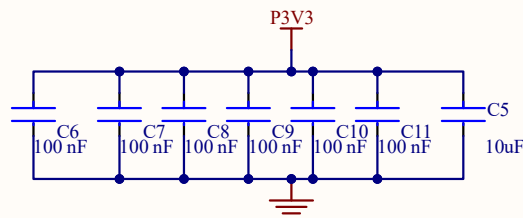
Note:
Route the USB nets with
90 ohm +/- 15 % differential impedance



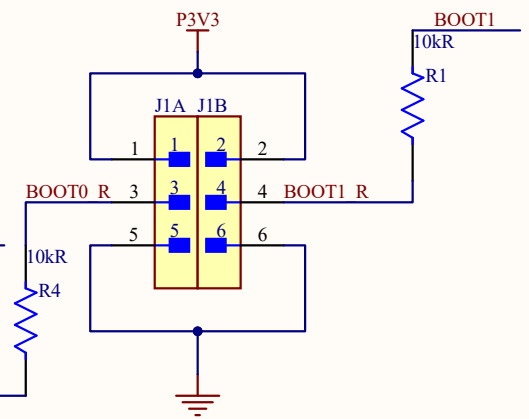
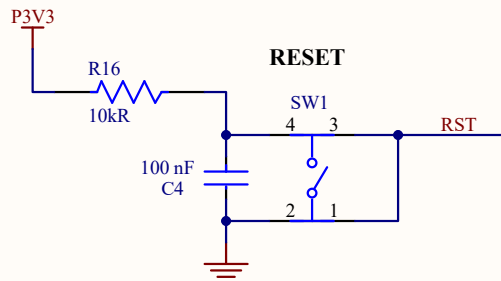
TEMP ALERT LED



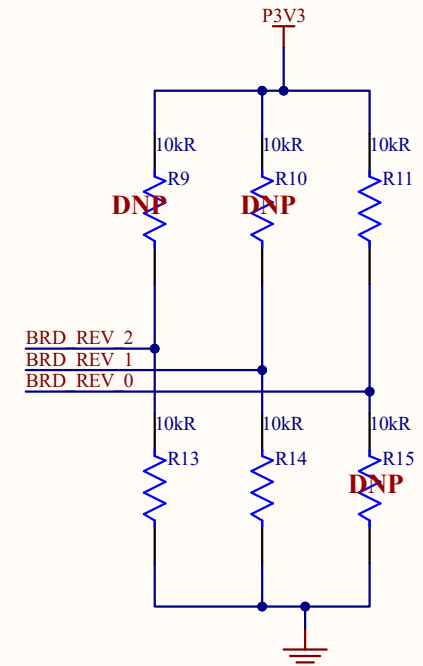
DECOUPLING CAPACITORS



RESET



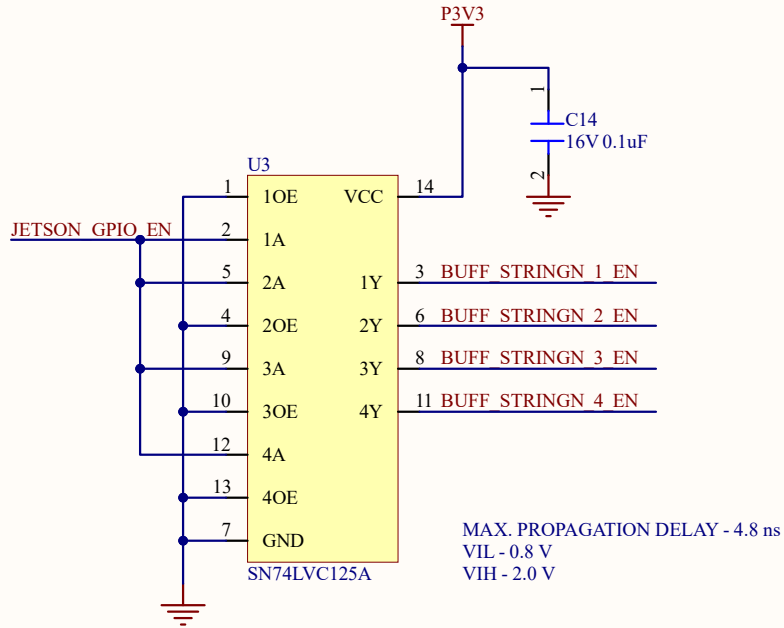
BOOT mode selection pins		Boot mode
BOOT1	BOOT0	
x	0	Main flash memory
0	1	System memory
1	1	Embedded SRAM



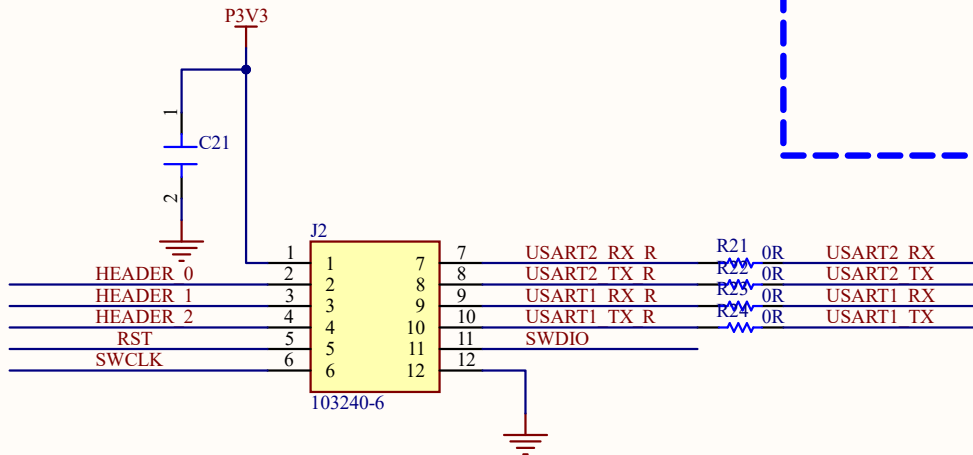
BOARD REV: 001

Title		
LED DRIVING MODULE		
Size	Number	Revision
A		V1.0
Date:	6/14/2026	Sheet of
File:	C:\Users\...\MICROCONTROLLER.Sch	Drawn By: GADILINGAREDDY G M

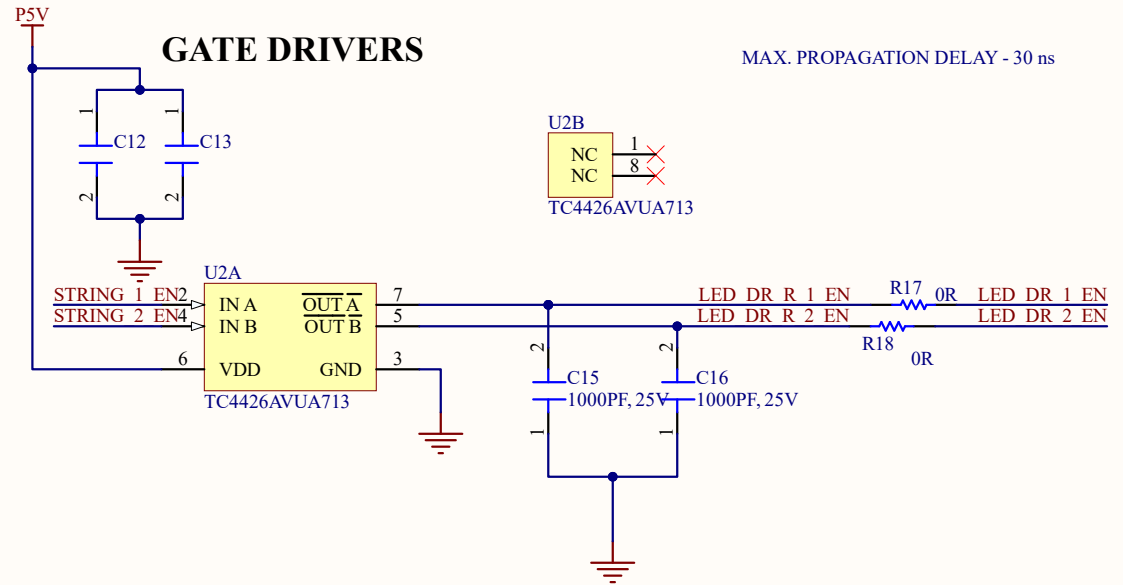
BUFFER



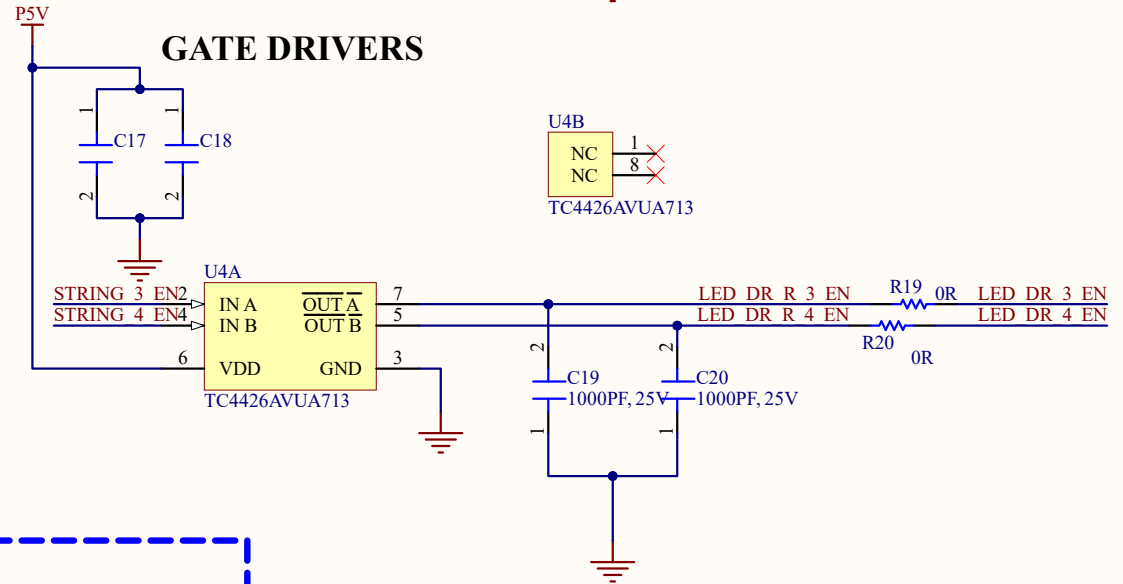
HEADERS



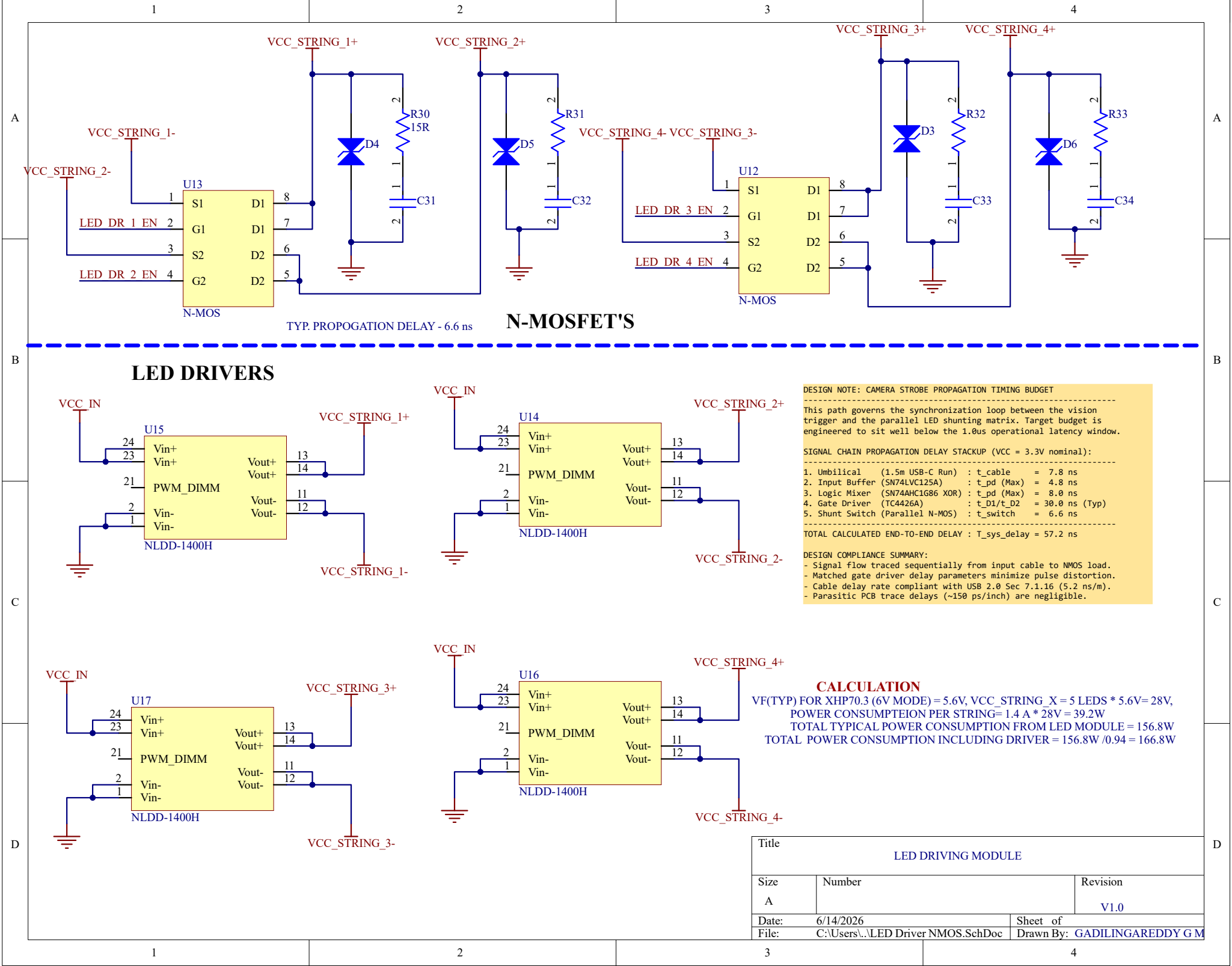
GATE DRIVERS



GATE DRIVERS



Title		
LED DRIVING MODULE		
Size	Number	Revision
A		V1.0
Date:	6/14/2026	Sheet of
File:	C:\Users\...\GATE_DRV_R_BFR_HDR.Sch	Drawn By: GADILINGAREDDY G M



LED DRIVERS

DESIGN NOTE: CAMERA STROBE PROPAGATION TIMING BUDGET

This path governs the synchronization loop between the vision trigger and the parallel LED shunting matrix. Target budget is engineered to sit well below the 1.0us operational latency window.

SIGNAL CHAIN PROPAGATION DELAY STACKUP (VCC = 3.3V nominal):

- 1. Umbilical (1.5m USB-C Run) : t_cable = 7.8 ns
- 2. Input Buffer (SN74LVC125A) : t_pd (Max) = 4.8 ns
- 3. Logic Mixer (SN74AHC1686 XOR) : t_pd (Max) = 8.0 ns
- 4. Gate Driver (TC4426A) : t_D1/t_D2 = 30.0 ns (Typ)
- 5. Shunt Switch (Parallel N-MOS) : t_switch = 6.6 ns

TOTAL CALCULATED END-TO-END DELAY : T_sys_delay = 57.2 ns

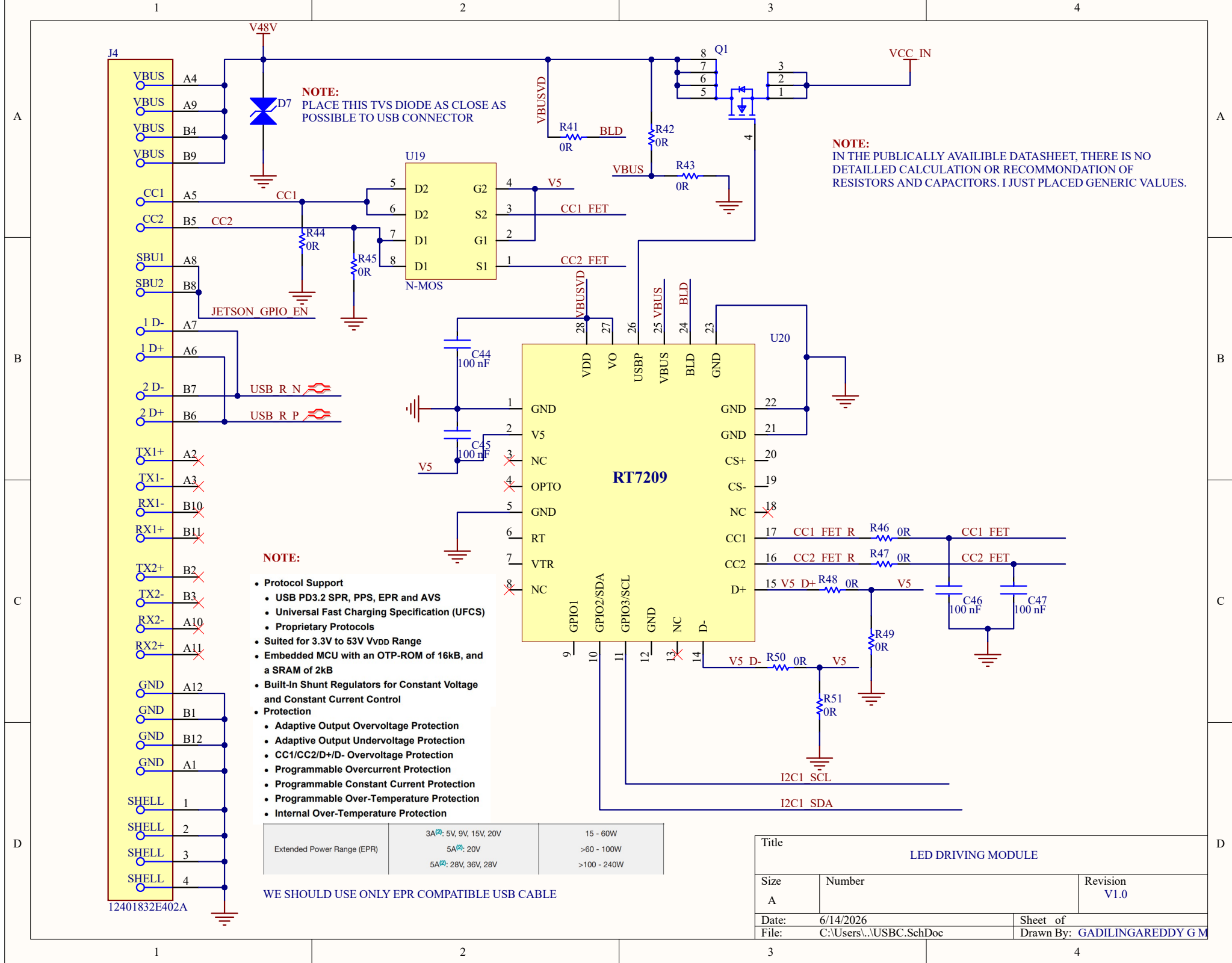
DESIGN COMPLIANCE SUMMARY:

- Signal flow traced sequentially from input cable to NMOS load.
- Matched gate driver delay parameters minimize pulse distortion.
- Cable delay rate compliant with USB 2.0 Sec 7.1.16 (5.2 ns/m).
- Parasitic PCB trace delays (~150 ps/inch) are negligible.

CALCULATION

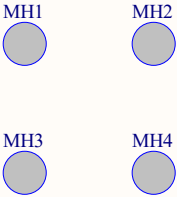
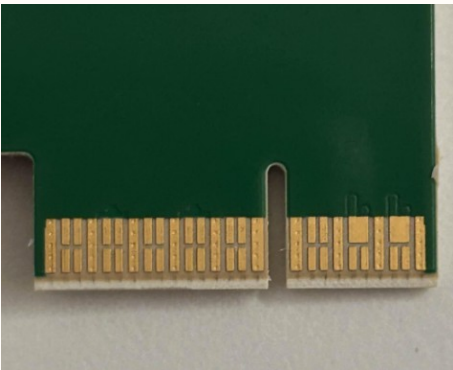
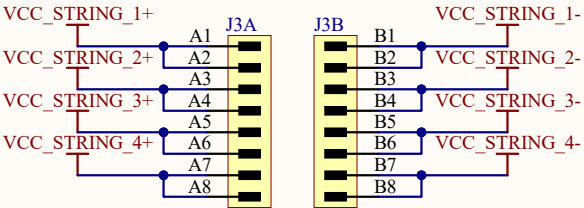
VF(TYP) FOR XHP70.3 (6V MODE) = 5.6V, VCC_STRING_X = 5 LEDS * 5.6V = 28V,
POWER CONSUMPTION PER STRING = 1.4 A * 28V = 39.2W
TOTAL TYPICAL POWER CONSUMPTION FROM LED MODULE = 156.8W
TOTAL POWER CONSUMPTION INCLUDING DRIVER = 156.8W / 0.94 = 166.8W

Title			LED DRIVING MODULE		
Size	Number		Revision		
A			V1.0		
Date:	6/14/2026			Sheet of	
File:	C:\Users\...\LED Driver NMOS.SchDoc			Drawn By:	GADILINGAREDDY G M

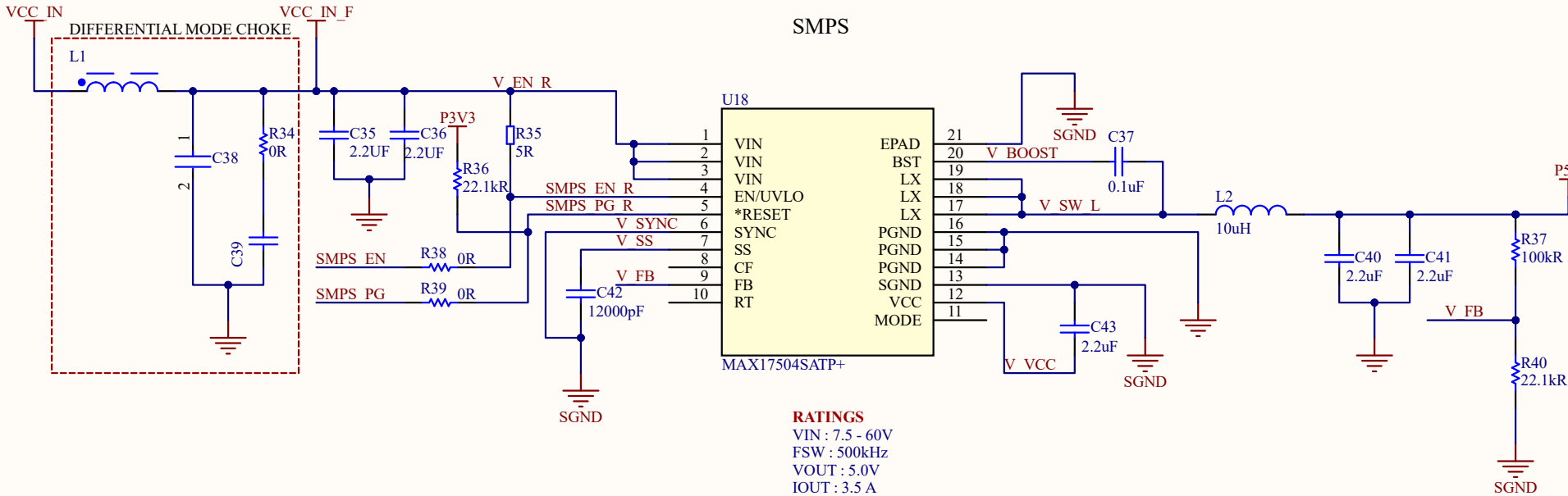


EDGE CONNECTOR FOR LED MODULE

MOUNTING HOLES



SMPS



Title		
LED DRIVER MODULE		
Size	Number	Revision
A		V1.0
Date:	6/14/2026	Sheet of
File:	C:\Users\...\SMPS PROTECTION.SchDoc	Drawn By: GADILINGAREDDY G M